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LAB MANUAL
Ethical Hacking Laboratory – Kali Linux Commands and Nmap Scanning
Course Code: Ethical Hacking Laboratory
Software Required: Kali Linux (Latest Version

Experiment 8

Nmap Service Version and OS Detection

Aim
To perform service version detection and operating system identification using Nmap.
Theory

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing.

Commands
i) Detect Service Versions
nmap -sV 192.168.1.10
Description:
Determines the versions of services running on target ports.

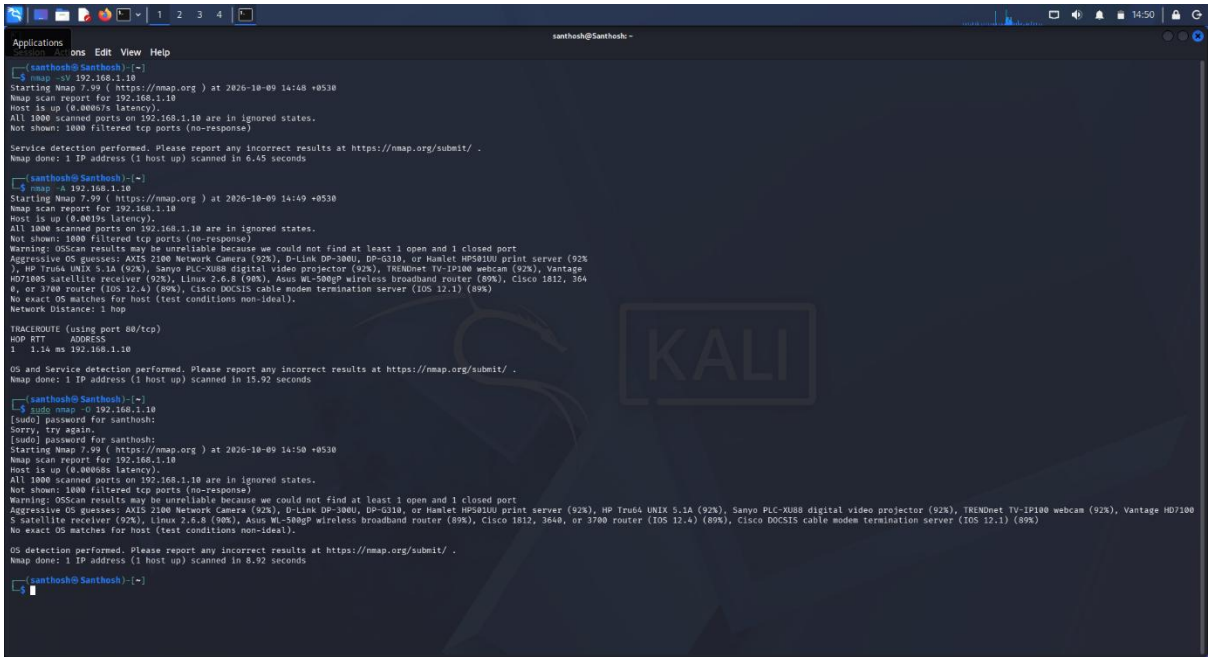
ii) Aggressive Scan
nmap -A 192.168.1.10
Description:
Performs OS detection, version detection, script scanning, and traceroute.

iii) Operating System Detection
sudo nmap -O 192.168.1.10

Description:
Identifies the operating system running on the target host.

Sample Output
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 8.2
80/tcp open http Apache 2.4

Result
Successfully performed service version detection and OS identification using Nmap.


Applications
Edit View Help
santhosh@Santhosh: ~
14:50
┌──(santhosh㉿Santhosh)-[~]
└─$ nmap -sV 192.168.1.10
Starting Nmap 7.99 (https://nmap.org) at 2026-10-09 14:48 +0530
Nmap scan report for 192.168.1.10
Host is up (0.00067s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.45 seconds
┌──(santhosh㉿Santhosh)-[~]
└─$ nmap -A 192.168.1.10
Starting Nmap 7.99 (https://nmap.org) at 2026-10-09 14:49 +0530
Nmap scan report for 192.168.1.10
Host is up (0.0019s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%
), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage
HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 364
0, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.14 ms 192.168.1.10
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.92 seconds
┌──(santhosh㉿Santhosh)-[~]
└─$ sudo nmap -O 192.168.1.10
[sudo] password for santhosh:
Sorry, try again.
[sudo] password for santhosh:
Starting Nmap 7.99 (https://nmap.org) at 2026-10-09 14:50 +0530
Nmap scan report for 192.168.1.10
Host is up (0.00056s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100
S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.92 seconds
┌──(santhosh㉿Santhosh)-[~]
└─$